

0923 U1 experiment

1. Publishers of a magazine wish to determine what proportion of the magazine's 50,000 subscribers are pleased with their subscription. The publishers intend to mail a survey to 1,000 subscribers randomly selected from those who have received the magazine for 5 years or more. This introduces selection bias, since long-subscribing customers are more likely to be pleased with their subscription. Which of the following would best eliminate selection bias?
- (A) Mail surveys to 2,000 subscribers randomly selected from those who have received the magazine for 5 years or more.
 - (B) Mail surveys to 1,000 subscribers randomly selected from those who have received the magazine for 1 year or less.
 - (C) Mail surveys to 1,000 subscribers randomly selected from all subscribers. ✓
 - (D) Mail surveys to 500 subscribers randomly selected from a group who have received a free six-month subscription within the past year.
2. A survey was administered to parents of high school students in a certain state to see if the parents thought the students' academic needs were being met. To select the sample, the parents were divided into two groups—one group of parents who live in cities with populations of more than 100,000 and the other group of parents who live in cities with populations less than or equal to 100,000. A random sample of 100 parents from each group was taken. Which of the following statements about the sample of 200 parents is true?
- (A) It is a stratified random sample because parents were randomly selected from each group. ✓
 - (B) It is a random cluster sample because parents were randomly selected from each group.
 - (C) It is a random cluster sample because groups of high schools were randomly selected.
 - (D) It is a systematic sample because the parents were systematically divided into two groups.
3. The buyer for an electronics store wants to estimate the proportion of defective wireless game controllers in a shipment of 5,000 controllers from the store's primary supplier. The shipment consists of 200 boxes each containing 25 controllers. The buyer numbers the boxes from 1 to 200 and randomly selects six numbers in that range. She then opens the six boxes with the corresponding numbers, examines all 25 controllers in each of these boxes, and determines the proportion of the 150 controllers that are defective. What type of sample is this?
- (A) Biased random sample
 - (B) Simple random sample
 - (C) Stratified random sample
 - (D) Cluster random sample ✓

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4. A city has designed a survey to collect information about residents' opinions about city services. Which of the following describes a scenario in which nonresponse bias is likely present?

(A) Surveys were mailed to 500 people, and 200 of the surveys were completed and returned. ✓

(B) Surveys were mailed to one-half of the neighborhoods in the city and all surveys were returned

(C) Surveys were completed by the first 500 residents who listened to a radio show

(D) Surveys were completed by 500 residents selected at random

Answer A

Correct. Since only 200 surveys out of the 500 surveys mailed were completed and returned, this high nonresponse rate is likely to lead to nonresponse bias.

5. A randomized block design will be used in an experiment to compare two lotions that protect people from getting sunburned. Which of the following should guide the formation of the blocks?

(A) Participants in the same block should receive the same lotion.

(B) Participants should be randomly assigned to the blocks.

(C) Participants within each block should be as similar as possible with respect to how easily they get sunburned. ✓

(D) Participants within each block should be as different as possible with respect to how easily they get sunburned.

6. A florist wanted to investigate whether a new powder added to the water of cut flowers helps to keep the flowers fresh longer than just water alone. For a shipment of roses that was delivered to the store, the florist flipped a coin before placing each rose in its own individual container with water. If the coin landed heads up, the rose was placed in water with the new powder; otherwise, the rose was placed in water alone. Which of the following is the best description of the method used by the florist?

(A) A census, because all roses are assigned to a container

(B) An experiment with a completely randomized design ✓

(C) An experiment with a blocked design, with blocking by type of water

(D) An experiment with a matched-pairs design

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7. The quality-control manager of a large factory is concerned about the number of defective items produced by workers. Thirty workers at the factory agree to participate in a study of three different incentive plans to help reduce the number of defective items produced. The plans will be randomly assigned to the workers so that 10 workers received each plan. The reduction in the number of defective items produced by each worker will be recorded two weeks after the plans are implemented.

Which of the following best describes why a completely randomized design is an appropriate design to use in this situation?

- (A) There is no blocking variable, and incentive plans will be randomly assigned to the workers. ✓
- (B) There is no blocking variable, and the workers were selected at random.
- (C) Each incentive plan is a block, and a completely randomized design is not blocked.
- (D) Each plan will be randomly assigned to 10 pairs of workers who share a similar characteristic.

Answer A

Correct. The treatments, which are the incentive plans, will be randomly assigned to the 30 workers.

8. Which of the following can be used to show a cause-and-effect relationship between two variables?

- (A) A census
- (B) A controlled experiment ✓
- (C) An observational study
- (D) A sample survey

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9. A medical researcher conducted a study to investigate the effectiveness of two drugs, L and M, designed to reduce the number of seizures that people with epilepsy might have. Each person in a group of 120 adult volunteers who had epilepsy was randomly assigned to one of two groups. Those assigned to one group took drug L for one year, and those assigned to the other group took drug M for one year. Each participant recorded the number of seizures that occurred during the year.

To which group is it most reasonable to generalize the results of this study?

- (A) All adults with epilepsy
- (B) All adults with epilepsy who take drugs to treat the disease
- (C) All adults with epilepsy who are similar to those who volunteered for the study ✓
- (D) All adults who experience a reduction in the number of seizures

Answer C

Correct. The largest population to which the results of a study can be generalized is the population from which the sample was selected. Because the study only uses a group of volunteers and not a random sample, the results can only be generalized to those similar to the volunteers.

10. An experiment will be conducted to determine whether children learn their multiplication facts better by practicing with flash cards or by practicing on a computer. Children who volunteer for the experiment will be randomly assigned to one of the two treatments. Because the children's gender may affect the outcome, there will be blocking by gender. After practice, the children will be given a test on their multiplication facts. Why will it be impossible to conduct a double-blind experiment?
- (A) The experimenter will know whether the child is a boy or a girl and whether he or she used flash cards or the computer.
 - (B) The child will know whether he or she is a boy or a girl.
 - (C) The child will know whether he or she used flash cards or the computer. ✓
 - (D) The person who grades the tests will know whether the child was a boy or a girl.

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11. To investigate whether the consumption of beetroot juice enhances exercise performance, a researcher selected a random sample of 50 student athletes from all the student athletes at a college. The athletes in the sample were randomly assigned to one of two groups. In one group, 25 athletes were given a daily dose of beetroot juice, and in the other group, the remaining athletes were given a daily dose of a placebo. At the end of six weeks of exercise training, the researcher compared the performances of the two groups.

Based on the design of the investigation, which of the following is the largest population to which the results can be generalized?

- (A) The 25 student athletes assigned to the beetroot juice group
- (B) The 50 student athletes in the sample
- (C) All student athletes at the college
- (D) All people who exercise

**Answer C**

Correct. The largest population to which the results can be generalized is the population from which the sample was selected, which is all student athletes at the college.

12. To investigate the effectiveness of an herbal medication for pain relief, a researcher randomly selected 80 patients who were undergoing physical therapy while recovering from an injury. Of the patients, 40 had knee injuries, 16 had arm injuries, and 24 had foot injuries. The researcher will use two treatments for the investigation: the herbal medication and a standard medication. After the patients are on the medication for two weeks, the amount of pain reduction will be measured for each patient. Which of the following best describes why a randomized block design is appropriate for the investigation?

- (A) The 80 patients were selected at random from all patients undergoing physical therapy while recovering from an injury.
- (B) Each patient will be randomly assigned to either the herbal medication or the standard medication.
- (C) Blocking on injury type will help to separate natural variability from differences due to injury type.
- (D) Each patient assigned to the herbal medication will be paired with a patient assigned to the standard medication.

**Answer C**

Correct. The type of injury is a blocking variable, and treatments will be randomly assigned within the blocks. Blocking ensures that units within each block are similar with respect to the blocking variable, and helps to separate natural variability due to the blocking variable.

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13. A pottery school wants to study whether the glaze of stoneware bowls is affected by the temperature used to harden the bowls. The study will use 3 different types of clay because glaze can affect different clay in different ways. Eight bowls will be made with each clay type for a total of 24 bowls, and 4 different temperatures will be used. What is the best design to use for the study?
- (A) A completely randomized design. Randomly assign temperatures so that 2 bowls of each clay type are glazed at each temperature.
 - (B) A randomized block design. Randomly assign temperatures so that 6 bowls are glazed at each temperature.
 - (C) A randomized block design. Randomly assign temperatures so that 2 bowls of each clay type are glazed at each temperature. ✓
 - (D) A matched pairs design. Randomly assign temperatures so that 2 bowls of each clay type are glazed at each temperature.

Answer C

Correct. Because the clay differs in ways that affect glaze quality, blocking on clay type is better than a completely randomized design. This way, all clay types are tested at all temperatures.

14. A local employer asked for help selecting a new type of desk chair. Thirty employees volunteered, and each employee used the new desk chair for two weeks and the current desk chair for two weeks. To determine which chair was used first, a coin was flipped for each employee. Heads represented using the new chair first, and tails represented using the current chair first. At the end of each two-week period, the employees were asked to rate their satisfaction with the new chair. Which of the following best describes this study?
- (A) It is not a well-designed experiment because employees were allowed to volunteer instead of being randomly selected.
 - (B) It is not a well-designed experiment because each employee was assigned both treatments, and each employee should be assigned only one.
 - (C) It is not a well-designed experiment because using a coin flip does not guarantee that fifteen employees will use each chair first.
 - (D) It is a well-designed experiment because there is random assignment, replication, and comparison of at least two treatment groups. ✓

Answer D

Correct. Random assignment, replication, and the ability to compare at least two treatment groups are all criteria for a well-designed experiment.

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15. A paint manufacturer conducted an experiment to investigate whether a new formula produces paint that lasts longer on the exterior of a building than paint produced using an older formula. A grid of 20 squares was drawn on one side of an exterior wall of a building. The new formula was randomly assigned to 10 squares, and the older formula was assigned to the remaining 10 squares. Each square was painted with the assigned formula. One year later, each square was rated on how well the paint had lasted, using a numerical scale for which larger values indicated longer-lasting paint.

Which of the following best describes the design of the experiment?

- (A) A completely randomized design, because each formula was randomly assigned to 10 squares. ✓
- (B) A matched pairs design, because the same number of squares were assigned to each formula.
- (C) A matched pairs design, because two formulas were compared.
- (D) A randomized block design, because the wall was partitioned into 20 squares.

Answer A

Correct. In a completely randomized design, treatments (new and older formulas of paint) are assigned to experimental units (squares on one side of an exterior wall of a building) completely at random.

16. A dog food company wishes to test a new high-protein formula for puppy food to determine whether it promotes faster weight gain than the existing formula for that puppy food. Puppies participating in an experiment will be weighed at weaning (when they begin to eat puppy food) and will be weighed at one-month intervals for one year. In designing this experiment, the investigators wish to reduce the variability due to natural differences in puppy growth rates. Which of the following strategies is most appropriate for accomplishing this?

- (A) Block on dog breed and randomly assign puppies to existing and new formula groups within each breed. ✓
- (B) Block on geographic location and randomly assign puppies to existing and new formula groups within each geographic area.
- (C) Stratify on dog breed and randomly sample puppies within each breed. Then assign puppies by breed to either the existing or the new formula.
- (D) Stratify on geographic location of the puppies and randomly sample puppies within each geographic area. Then assign puppies by geographic area to either the existing or the new formula.

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17. The owner of a plant nursery will conduct a study to investigate whether a new fertilizer is more effective than an older fertilizer in helping the growth of young tomato seedlings. The owner selects a simple random sample of 50 young tomato seedlings from all the young tomato seedlings in the nursery. Half of the selected seedlings will be randomly assigned to receive the new fertilizer, and the remaining seedlings will receive the older fertilizer. The growth of the seedlings will be recorded for the next month. What is the main reason in the study for selecting a simple random sample from the population?
- (A) A simple random sample should be used to obtain all experimental units.
 - (B) A simple random sample allows for the random assignment of treatments to experimental units.
 - (C) The simple random sample allows the researcher to establish a causal relationship among the treatments.
 - (D) A simple random sample selected from the population allows the results to be generalized to the population. ✓

Answer D

Correct. A simple random sample is likely to be representative of the population and will allow results to be generalized to the population.

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18. A researcher randomly selected 30 people to participate in a study to investigate which of two materials, A or B, used in the heels of sneakers will last longer. The researcher is considering two design types for the study.
- I. Design 1: Randomly assign 15 participants to wear sneakers with material A on their left feet and material B on their right feet. The remaining participants will wear sneakers with material B on their left feet and material A on their right feet.
 - II. Design 2: Randomly assign 15 participants to wear sneakers with material A on both feet and the remaining participants to wear sneakers with material B on both feet.

Which of the following best describes each design?

- (A) Design 1 is a completely randomized design, and design 2 is a matched pairs design.
- (B) Design 1 is a matched pairs design, and design 2 is a completely randomized design. ✓
- (C) Design 1 is a completely randomized design, and design 2 is a completely randomized design.
- (D) Design 1 is a completely randomized design, and design 2 is a randomized block with materials as blocks.

Answer B

Correct. Design 1 is a matched pairs design because each person is considered a pair and receives both types of materials, with the type of material randomly assigned to the left or right foot. Design 2 is a completely randomized design because the type of material is randomly assigned to each person.

19. Eighteen individuals who use a particular form of social media were assigned a new user interface to use when logging in to their accounts. After using the new user interface for a week, each individual was asked to rate how easy or hard the new user interface was to use on a scale from 1 (extremely easy) to 9 (extremely hard). Which of the following correctly identifies why this is not a well-designed experiment?
- (A) There was a lack of control because not all individuals in the study used login passwords of the same length.
 - (B) The individuals may not have been randomly selected.
 - (C) There was not enough replication because the individuals used the new user interface for only one week.
 - (D) The study was not comparative—only one treatment was used. ✓

Answer D

Correct. Well-designed experiments should involve comparisons of at least two treatment groups, one of which could be a control group.

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20. A matched-pairs t -test is NOT an appropriate way to analyze data consisting of which of the following?

- (A) Measurements of annual income taken both before and after a two-year training course for a random sample of 100 people who took the course
- (B) Measurements of annual income for each twin for 100 randomly selected pairs of twins
- (C) Measurements of annual income for both individuals in pairs formed by matching 100 people from State A and 100 people from State B based on level of education
- (D) Measurements of annual income for both individuals in pairs formed by assigning 100 people to pairs at random ✓